

# Risk Markups

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# Risk Markups, Misallocation, and Optimal Policy

- Large literature estimating sizable and heterogeneous markups (Hall; De Loecker et. al.)
- Markups can lower TFP, distort aggregate employment, and create inequality
- Role for policy in addressing markups/misallocation depends on source of markups
- E.g., common view: markups come from market power
  - Baqaee and Farhi (2020); Edmond, Midrigan and Xu (2023)
  - Optimal policy: increase output with subsidies
- Alternative view: measured markups reflect risk premiums
  - Knight (1921); Joel David et al. (2022, 23, 23); Boar, Gorea, and Midrigan (2023)
  - Optimal policy: ???
    - Are risk markups inefficient?

1. Model of risk markups arising from endogenously incomplete idiosyncratic risk-sharing
  - Start with simple model of entrepreneurship (like Moll 2014)
  - Remove capital/capital constraints
  - Add risk and incomplete risk sharing
  - Competitive markets, homogeneous good, linear production, no market power
2. Main Contribution: Constrained efficient allocation and implementation
  - Baseline case: policy relevant and useful to compare to market-power models
  - Rich-instrument case: saturated planner problem useful to understand inefficiency

# Main Result: Optimal to Tax Labor in Long Run

- Policy instruments:  $t_0$  transfers & uniform labor tax  $\tau_{\ell t}$
- In the long run, optimal policy equals:

$$1 + \tau_{\ell}^* = \underbrace{\frac{1}{1 + \pi^*}}_{\text{aggregate markup}} \times \underbrace{\frac{\gamma_w^*}{\eta_w^*}}_{\text{worker Pareto weight / worker consumption share}}$$

- In the long run, optimal policy is a tax (not subsidy)
- Inefficient markups vs. inefficient precautionary saving
- Inefficient risk markups: subsidy equal to markup<sup>-1</sup>;  $1 + \tau^* = \frac{1}{1 + \pi^*}$ , like market power
- Excess saving by entrepreneurs makes workers too poor, thus they supply too much labor

# The Model

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# Economic Environment

- Representative worker and entrepreneurs  $i \in [0, 1]$

$$U_w(c_w, \ell) = \mathbb{E} \left[ \int_0^\infty e^{-\rho_w t} \left( \log c_{wt} - \frac{\ell_t^{1+1/\psi}}{1+1/\psi} \right) dt \right]$$

$$U_e(c_i) = \mathbb{E} \left[ \int_0^\infty e^{-\rho_e t} \log c_{it} dt \right]$$

- Technology

$$dY_{it} = \ell_{it} (z_{it} dt + \sigma_{yit} dB_{yit}),$$

$$dx_{it} = \mu_x(x_{it}) dt + \sigma_x(x_{it}) dB_{xit},$$

$$z_{it} = z(x_{it}); \sigma_{yit} = \sigma_y(x_{it}).$$

- Resource constraints

$$c_t = \int_0^1 c_{it} di + c_{wt} = \int_0^1 z_{it} \ell_{it} di$$

$$\int_0^1 \ell_{it} di = \ell_t$$

# Competitive Equilibrium

- Competitive goods, bond, and labor market with wage  $w_t$  and interest rate  $r_t$
- Agents, preferences and technology as described in environment
- Representative worker chooses  $c_{wt}, \ell_t$ ; entrepreneurs choose  $c_{it}, \ell_{it}$
- Resource constraints / Market clearing

$$\int_0^1 c_{it} di + c_{wt} = \int_0^1 z_{it} \ell_{it} di; \quad \int_0^1 \ell_{it} di = \ell_t$$
$$n_{wt} + \int_0^1 n_{it} di = 0$$

- Budget constraints
  - Entrepreneurs can sell up to  $(1 - \phi(x_{it}))$  fraction of firm (and do)

$$dn_{wt} = (r_t n_{wt} - c_{wt} + w_t \ell_t) dt$$

$$dn_{it} = (r_t n_{it} - c_{it} + (z_{it} - w_t) \ell_{it}) dt + (\phi_{it} \sigma_{yit}) \ell_{it} dB_{yit}.$$

- Natural borrowing constraints

- Standard Euler equation and labor supply FOC:

$$\frac{dc_{wt}}{c_{wt}} = (r_t - \rho_w)dt$$

$$c_{wt}^{-1} w_t = \ell_t^{1/\psi}$$

# Entrepreneur Choices: Risk vs. Expected Return and Precautionary Savings

$$\frac{dc_{it}}{c_{it}} = (r_t - \rho_e + \sigma_{cit}^2)dt + \sigma_{cit}dB_{yit},$$

$$\sigma_{cit} = (\phi_{it}\sigma_{yit})\frac{\ell_{it}}{n_{it}}$$

$$z_{it} - w_t \leq \sigma_{cit}\phi_{it}\sigma_{yit} \quad (\text{with equality if } z_{it} \geq w_t)$$

- $\sigma_{cit}^2 dt$  in Euler equation is precautionary saving
- Choose scale  $\ell_{it}$  by comparing expected profit to risk; bonds are self-insurance:

$$\ell_{it} = \frac{(z_{it} - w_t)^+}{(\phi_{it}\sigma_{yit})^2} n_{it}$$

- Risk-adjusted mg. cost is equalized:  $z_{it} - \sigma_{cit}\phi_{it}\sigma_y = w_t$ , for  $z_{it} \geq w_t$

$$1 + \pi_{it} := \frac{1}{w_t/z_{it}} = 1 + \frac{\sigma_{cit}\phi\sigma_{yit}}{w_t}$$

- Markups in Hall-De Loecker estimation
- Markups depend on productivity relative to wage
- Even highly diversified firms could have large markup
- Diversification appears as large size

# Aggregation: TFP and Aggregate Markup

- TFP is a labor-weighted average of idiosyncratic productivity

$$Z_t := \frac{c_t}{\ell_t} = \int_0^1 z_{it} \left( \frac{w_t \ell_{it}}{w_t \ell_t} \right) di$$

- The aggregate markup is a cost-weighted average of idiosyncratic markups

$$\begin{aligned} 1 + \pi_t &:= \frac{Z_t}{w_t} = \int_0^1 \frac{z_{it}}{w_t} \left( \frac{w_t \ell_{it}}{w_t \ell_t} \right) di \\ &= \int_0^1 (1 + \pi_{it}) \left( \frac{w_t \ell_{it}}{w_t \ell_t} \right) di \end{aligned}$$

# Aggregation: a Separation Result

- Aggregation: policy functions are linear in net worth (or consumption), so we only need to keep track of aggregate consumption share by  $x$ :  $\eta_t(x)$

$$\eta_t(x) := \int \eta g_t(\eta, x) dx; \quad \eta_{et} = \int \eta_t(x) dx; \quad \eta_{wt} = 1 - \eta_{et}$$

- Goods and labor market clearing

$$\frac{1}{\rho_e \phi(x) \sigma_y(x)} \times \int_{w_t}^{\infty} z(x) \eta_t(x) \frac{z(x) - w_t(\eta_t)}{\phi(x) \sigma_y(x)} dx = 1$$

$$c_t = \left( \frac{1}{\rho_e \phi(x) \sigma_y(x)} \times \int_{w_t}^{\infty} \eta_t(x) \frac{z(x) - w_t(\eta_t)}{\phi(x) \sigma_y(x)} dz \right)^{-1} \ell_t = Z_t(\eta_t) \ell_t$$

$$\sigma_{ct}(x; \eta_t) = \frac{(z(x) - w(\eta_t))^+}{\phi(x) \sigma_y(x)}$$

# Aggregation: a Separation Result

- Goods and labor market clearing

$$\frac{1}{\rho_e \phi_{it} \sigma_{yit}} \times \int_{w_t}^{\infty} z \eta_t(z) \frac{z - w_t(\eta_t)}{\phi_{it} \sigma_{yit}} dz = 1$$

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$$\sigma_{ct}(Z_{it}; \eta_t) = \frac{(Z_{it} - w_t(\eta_t))^+}{\phi_{it} \sigma_{yit}}$$

- Euler eqs. give law of motion for  $\eta_t$  (KFE) that only depends on  $\eta_t$ :

$$\partial_t \eta_t(x) = \mathcal{L}_x^* \eta_t(x) + A(x; \eta_t),$$

where  $\mathcal{L}_x^*$  is the adjoint of  $\mathcal{L}_x$  and  $A(x; \eta_t)$  is the geometric drift of  $\eta_{it}$  times  $\eta_t(x)$ .

# Aggregation: a Separation Result

- Goods and labor market clearing

$$c_t = \left( \frac{1}{\rho_e \phi_{it} \sigma_{yit}} \times \int_{w_t}^{\infty} \eta_t(z) \frac{z - w_t(\eta_t)}{\phi_{it} \sigma_{yit}} dz \right)^{-1} \ell_t = Z_t(\eta_t) \ell_t$$

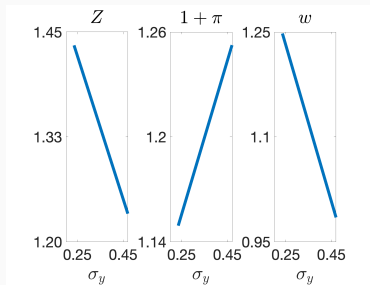
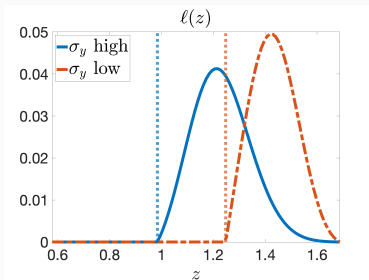
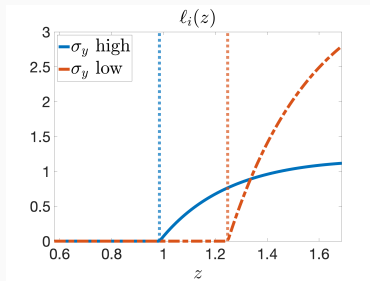
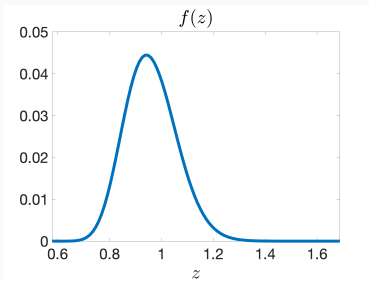
$$\frac{1}{\rho_e \phi_{it} \sigma_{yit}} \times \int_{w_t}^{\infty} z \eta_t(z) \frac{z - w_t(\eta_t)}{\phi_{it} \sigma_{yit}} dz = 1$$

$$\sigma_{ct}(z; \eta_t) = \frac{z - w_t(\eta_t)}{\phi_{it} \sigma_{yit}}$$

- Euler eqs. give law of motion for  $\eta_t$  (KFE)
- After pinning down the path of  $\eta_t(z)$ ,  $w_t$ ,  $\sigma_{ct}(z)$  and  $Z_t$ , we can use worker  $FOC(\ell)$  to solve for aggregate employment and output

$$\underbrace{((1 - \eta_{et}) Z_t \ell_t)}_{c_{wt}}^{-1} w_t = \ell_t^{1/\psi}$$

# Comparative Statics for $\sigma_y$



## **Constrained Efficiency + Implementation**

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# Constrained Efficiency: Two Approaches

## 1. Baseline

- Uniform labor tax (+ lump-sum transfers)
- Policy relevant and facilitates comparison to market-power

## 2. Rich-Instrument Case

- $x$ -targeted labor and savings taxes
- Saturated planner controls all margins
- Useful to understand inefficiency and baseline results
- Key Result: Same average labor tax formula as baseline case

# Planner Problem

- Standard Planner objective with Pareto weights  $\tilde{\Gamma}_w, \tilde{\Gamma}_e$  to choose  $(c_w, \ell, c_i, \ell_i)$

$$\int_0^\infty e^{-\rho_w t} \tilde{\Gamma}_w \left( \log c_{wt} - \frac{\ell_t^{1+1/\psi}}{1+1/\psi} \right) dt + \int_0^1 \mathbb{E} \left[ \int_0^\infty e^{-\rho_e t} \tilde{\Gamma}_e \log c_{it} dt \right] di$$

- Define normalized Pareto weights

$$\Gamma_{et} = e^{-\rho_e t} \tilde{\Gamma}_e, \quad \Gamma_{wt} = e^{-\rho_w t} \tilde{\Gamma}_w, \quad \gamma_{et} = \frac{\Gamma_{et}}{\Gamma_{wt} + \Gamma_{et}}, \quad \gamma_{wt} = \frac{\Gamma_{wt}}{\Gamma_{wt} + \Gamma_{et}}.$$

- Can be rewritten to separate labor  $\ell_t$  and distribution terms

$$\begin{aligned} & \overbrace{\int_0^\infty (\Gamma_{wt} + \Gamma_{et}) \log \ell_t - \Gamma_{wt} \frac{\ell_t^{1+1/\psi}}{1+1/\psi} dt}^{\text{labor}} + \overbrace{\int_0^\infty (\Gamma_{wt} + \Gamma_{et}) \log Z_t dt}^{\text{TFP}} \\ & + \underbrace{\int_0^\infty \Gamma_{wt} \log \eta_{wt} + \Gamma_{et} \left( \int p(x) \log \frac{\eta_t(x)}{p(x)} dx \right) dt}_{\text{workers vs. entrepreneurs}} - \underbrace{\int_0^\infty \Gamma_{et} \left( \int \left( p(x) \log \frac{\eta_t(x)}{p(x)} - \int \log \eta g_t(\eta, x) d\eta \right) dx \right) dt}_{\text{entrepreneur risk}} \end{aligned}$$

# First Best

Optimize each term in objective function:

$$\begin{aligned}\ell_t^{FB} &= \gamma_{wt}^{-\frac{\psi}{\psi+1}} \\ Z_t^{FB} &= \bar{z}, \\ \eta_{wt}^{FB} &= \gamma_{wt} \\ \eta_{it}^{FB} &= \frac{\eta_t^{FB}(x)}{p(x)} = \eta_{et}^{FB} = \gamma_{et}.\end{aligned}$$

- With complete markets, implement FB without taxes and only time-0 transfers
- Full insurance, so  $\sigma_{cit} = 0$ , so  $\eta_{it} = \eta_{et}$
- $w = \bar{z}$ , so  $Z = \bar{z}$  and labor supply is efficient because social cost=private cost=wage
- $\eta_{e0} = \frac{\Gamma_{e0}}{\Gamma_{e0} + \Gamma_{w0}}$

# Baseline Planner Problem

- Planner has uniform labor tax  $\tau_{\ell t}$ , rebated lump-sum to the worker
- Planner can redistribute at time-0 (or takes  $\eta_0(x)$  as constraint)
- $w_t = (1 + \tau_{\ell t})\tilde{w}_t$  is after-tax wage paid by entrepreneurs, and  $\tilde{w}_t$  the wage workers earn
- All equilibrium conditions the same except worker labor supply:

$$\ell_t^{1/\psi} = c_{wt}^{-1} \tilde{w}_t = c_{wt}^{-1} \frac{w_t}{1 + \tau_{\ell t}} = (\eta_{wt} Z_t \ell_t)^{-1} \frac{w_t}{1 + \tau_{\ell t}}$$

- Separation property of equilibrium
  - Planner can control  $\ell_t$
  - Choose  $\ell_t$  to maximize first term in objective
  - Choose  $\eta_x$  (if allowed) to maximize other distribution terms

# Optimal Labor and Implementation

- Constrained planner achieves first best labor:

$$\ell_t = [\eta_{wt} (1 + \pi_t) (1 + \tau_{\ell t})]^{-\frac{\psi}{1+\psi}} = \ell_t^{FB} = \gamma_{wt}^{-\frac{\psi}{\psi+1}}$$

- We can implement the optimal  $\ell_t$  with tax:

$$1 + \tau_{\ell t} = \frac{1}{1 + \pi_t} \times \frac{\gamma_{wt}}{\eta_{wt}}$$

- Risk markups reflect private cost of risk, but not social cost
  - Subsidy to offset markup, like market-power, but different reason
- Consumption distribution crucial for efficiency
  - Workers supply more labor when their consumption is lower (income effect)
  - If consumption is below first-best ( $\eta_{wt}^{FB} = \gamma_{wt}$ ), then need labor tax to implement FB labor
  - $\eta_{wt} \neq \gamma_{wt}$  due to entrepreneur precautionary saving

# Understanding the Labor Tax Result: Inefficient Risk Markups

$$1 + \tau_{\ell t} = \frac{1}{1 + \pi_t} \times \frac{\gamma_{wt}}{\eta_{wt}}$$

- Inefficient risk premium  $(1 + \pi_t)$ :

$$\sigma_{cit} = \frac{\phi_{it}\sigma_{yit}}{n_{it}}\ell_{it} = \rho_e \frac{\phi_{it}\sigma_{yit}}{\eta_{it}} \frac{1}{Z_t} \frac{\ell_{it}}{\ell_t}$$
$$n_{it} = \frac{1}{\rho_e} Z_t \ell_t \eta_{it}$$

- Entrepreneur takes  $n_{it}$  as given. Individual increase in  $\ell_{it}$  increases consumption volatility through increased exposure to risk
- Consider a planner who increases all  $\ell_{it}$  by constant proportion  $\rightarrow$  constant  $\sigma_{cit}$
- Increase in aggregate output  $\rightarrow$  increases net worth of entrepreneurs  $\rightarrow$  increases their insurance against increased risk  $\rightarrow$  no increase in consumption volatility
- (Past interest rates increase, by Euler equations, and entrepreneurs enter  $t$  with higher  $n_{it}$ )
- To correct inefficient risk premium: labor subsidy = aggregate markup<sup>-1</sup>
- As before, if  $\eta_{wt} \neq \gamma_{wt}$  need to offset income effect

# It is Optimal to Tax Labor, Not Subsidize

- For exposition, let  $\rho_e > \rho_w$ . Consider the long run  $\eta_t \rightarrow \eta^*$  and  $\gamma_{wt} \rightarrow 1$ , so

$$1 + \tau_{\ell t} = \zeta_t \times \gamma_{wt},$$

where

$$\zeta_t := \frac{w_t \ell_t}{c_{wt}}.$$

- Optimal policy in the long run in terms of income distribution:

$$1 + \tau_{\ell}^* = \frac{1}{1 + \pi^*} \times \frac{1}{\eta_w^*} = \zeta^* > 1$$

- Workers are paying interest on their debt to entrepreneurs, so they consume less than their labor income, so optimal long-run policy is a labor tax
- Equivalently, entrepreneurs are accumulating profits and saving precautionarily, so they have more wealth and consumption in the long-run than in FB
- If  $\Gamma_e$  not too large, optimal to tax on whole transition path

## Sufficient Statistics: Steady State

- Rewrite  $\ell_{\tau_\ell}^* = \left(\frac{1+\tau_\ell^*}{\zeta^*}\right)^{-\frac{\psi}{1+\psi}}$
- Recall  $\zeta^*$  invariant to tax. In long-run, only first term of planner objective affected by tax
- Welfare (first-term) of tax  $\tau_\ell^*$

$$\begin{aligned} v(\tau_\ell^*) &:= \log \ell_{\tau_\ell}^* - \frac{\psi}{1+\psi} (\ell_{\tau_\ell}^*)^{\frac{1+\psi}{\psi}} \\ &= -\frac{\psi}{1+\psi} \left( \log[1 + \tau_\ell^*] - \log \zeta^* + \frac{\zeta^*}{1 + \tau_\ell^*} \right). \end{aligned}$$

- Let  $\Delta v(\tau_\ell^*)$  be the consumption equivalent welfare gain of tax  $\tau_\ell^*$

$$\begin{aligned} \log[1 + \Delta v(\tau_\ell^*)] &= v(\tau_\ell^*) - v(0) \\ &= -\frac{\psi}{1+\psi} \left( \log[1 + \tau_\ell^*] - \zeta^* \frac{\tau_\ell^*}{1 + \tau_\ell^*} \right) \end{aligned}$$

- Consider 20% or 30% markups. Frisch labor supply elasticity of 1 or  
 $3.c_w \in \{0.85w\ell, 0.9w\ell, 0.95w\ell\}$

# Sufficient Statistics Welfare Analysis

- Optimal tax:

$$\log [1 + \Delta v(\zeta^*)] = -\frac{\psi}{1 + \psi} (\log \zeta^* - (\zeta^* - 1))$$

- Welfare gains range from 0.07% to 1.05%
- Mistaken tax to exactly offset markup:

$$\log \left[ 1 + \Delta v \left( \frac{1}{1 + \pi^*} \right) \right] = -\frac{\psi}{1 + \psi} (-\log [1 + \pi^*] + \zeta^* \pi^*)$$

- Welfare gain is negative and ranges from -6.57% to -1.40%
- Welfare gains along the transition path (full model calibration):
  - Steady-state to steady-state (no transition) welfare gain is 0.117%
  - Dynamic optimal policy welfare gain is 0.092%
  - Constant tax equal to the long-run optimal value welfare gain is 0.087%

## Summary of Baseline Results

- When markups reflect market power, optimal to subsidize
- When markups reflect risk compensation, optimal to tax (in long-run)
- Both cases want to implement first-best employment
- Risk markups are socially inefficient and reduce labor demand
- Risk induces entrepreneur precautionary saving, makes workers relatively poor, income effect increases labor supply
- Single feature of uninsurable idiosyncratic risk produces not only markups but also distortions in saving behavior. Consumption and wealth distribution crucial consideration for policy
- Simple welfare calculation suggests the welfare gain from the optimal tax is modest, while the welfare cost of a subsidy that offsets markups is large

# Saturated-Instruments Planner Problem

- The optimal tax depends on the consumption distribution, which is distorted from first best by entrepreneurs' precautionary saving
- If precautionary saving is inefficient, why doesn't the planner use entrepreneur-specific taxes on savings to undo it?
- How would access to savings taxes affect the optimal use of the labor tax?
- With Saturated-Instruments Planner Problem we have that:
  - Precautionary saving is inefficient, but it is not efficient to set consumption shares to first best
  - The labor tax formula is unchanged with a component that captures the deviation in consumption shares from first best
  - Better understand the source of inefficiency and how the planner would use targeted taxes to reduce misallocation

# Saturated-Instruments Planner Problem

- Planner can control labor  $\ell_t$ , labor share  $\hat{\ell}_t(x)$ , and the consumption distribution  $\eta_t(x)$
- Planner cannot complete markets, so still faces risk constraint (more labor, more risk)

$$\sigma_{ct}(x) = \frac{\rho_e \phi(x) \sigma_y(x)}{Z_t \eta_t(x)} \hat{\ell}_t(x)$$

- Details in paper for allocations and implementation.
  - $\ell_t = \ell_t^{FB}$ ;  $\hat{\ell}_t(x)$  increasing in productivity to improve TFP

$$\frac{\eta_t(x)}{p(x)} = \frac{\gamma_{et} \left( 1 + \frac{1}{\rho_e} \sigma_{ct}^2(x) \right)}{\gamma_{wt} + \gamma_{et} \left( 1 + \frac{1}{\rho_e} \sigma_{ct}^2(x) \right)}$$

- Planner distorts consumption and labor distributions relative to FB to improve risk sharing
- Average labor tax is same formula as Baseline case!

# Understanding the Saving Tax: Precautionary Saving

- In baseline,  $\eta_t$  not a control and  $\eta_t \neq \eta_t^{FB}$  from precautionary saving  $\sigma_{cit}^2 > 0$
- Even when  $\eta_t$  is a control,  $\eta_t \neq \eta_t^{FB}$
- Planner wants inverse Euler equation, entrepreneurs follow Euler equation
- Social optimality requires marginal cost of delivering utility to be a martingale. In standard settings, marginal cost is inverse of marginal utility of consumption, yielding inverse Euler equation
- In complete markets, Euler and inverse Euler equations align. Incomplete markets precautionary saving creates wedge
- Here, optimal tax has term to offset precautionary saving (which would implement first best) but also has term to give entrepreneurs more wealth because they use wealth to insure against risk.
- Either because Baseline  $\eta_t(x)$  not controlled or in rich-instrument case,  $\eta_t \neq \eta_t^{FB}$  so consumption distribution appears in labor tax

# Takeaways

1. Incomplete risk sharing creates risk markups, but also precautionary saving
2. Risk markups are inefficient
3. Precautionary saving is inefficient
4. Two inefficiencies call for opposite sign of labor tax (in isolation)
5. If measured markups reflect risk premia, an inefficient consumption, or wealth, distribution is an important consideration when designing optimal policy to address markups
  - Impoverished workers supply too much labor
6. Speculative: maybe optimal labor tax is close to zero ( $\frac{c_w}{w\ell} \approx 1$ ), so emphasis is not necessarily tax vs. subsidy, but rather caution in favor of zero intervention instead of big subsidy